

NirogScan Health Report - Dual Sample Analysis

Patient Information

Name: ashfaque
Age: 24
Gender: Male
Weight: 67
Height: N/A
Notes: 1min+

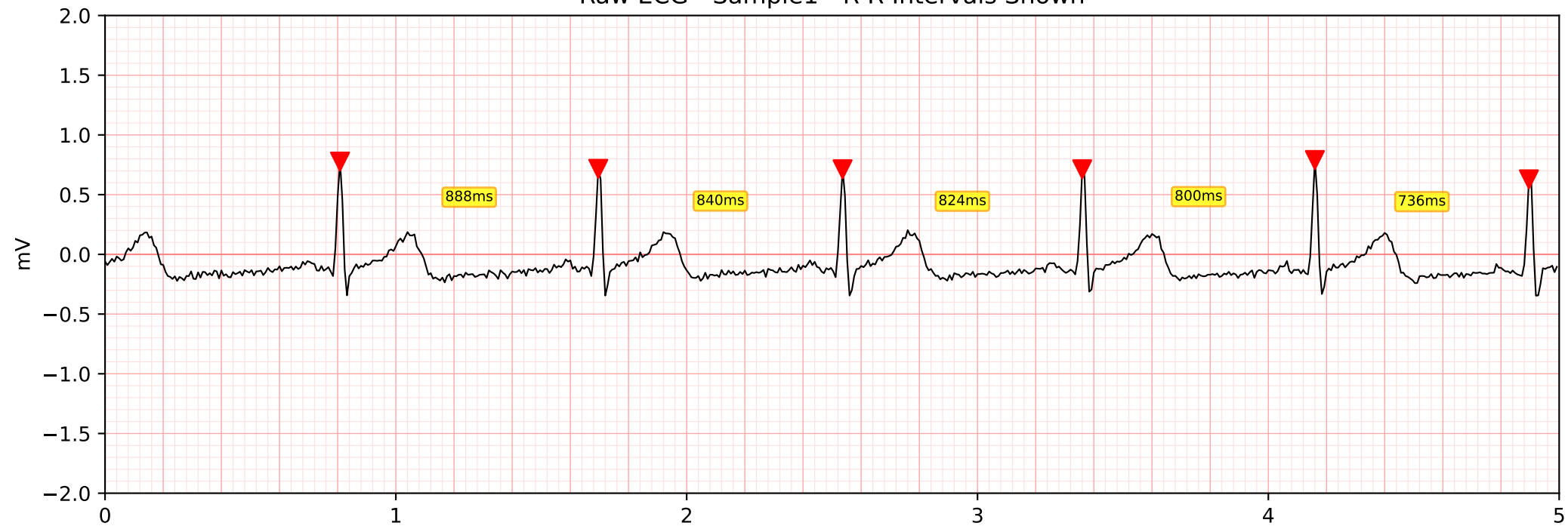
Analysis Summary

Total ECG Duration: 98.1s
ECG Sample 1: 5.0s (Best 5s Window - Basic Analysis)
ECG Sample 2: 98.1s (Full Recording - Complete Analysis)
Total PPG Duration: 196.2s
PPG Sample 1: Same as above (Basic Analysis)
PPG Sample 2: Same as above (Complete Long-term Analysis)

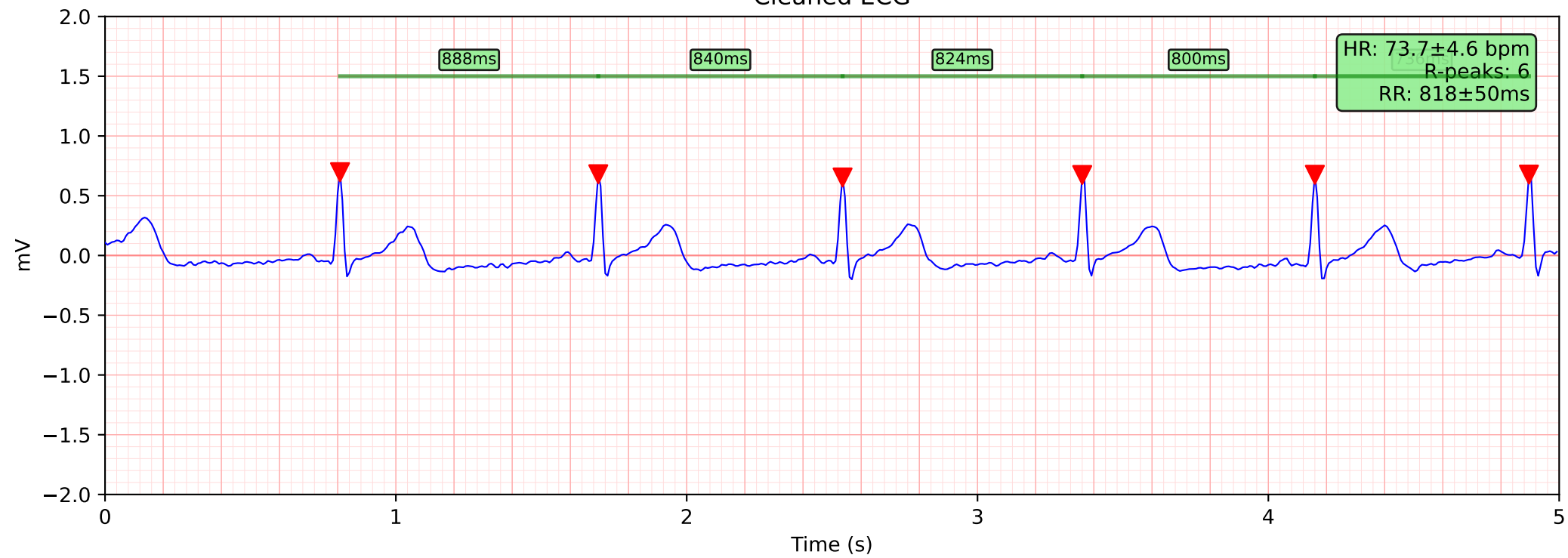
Generated: 2026-02-02 14:39:29

ECG - Sample 1: Best Quality

Raw ECG - Sample1 - R-R Intervals Shown

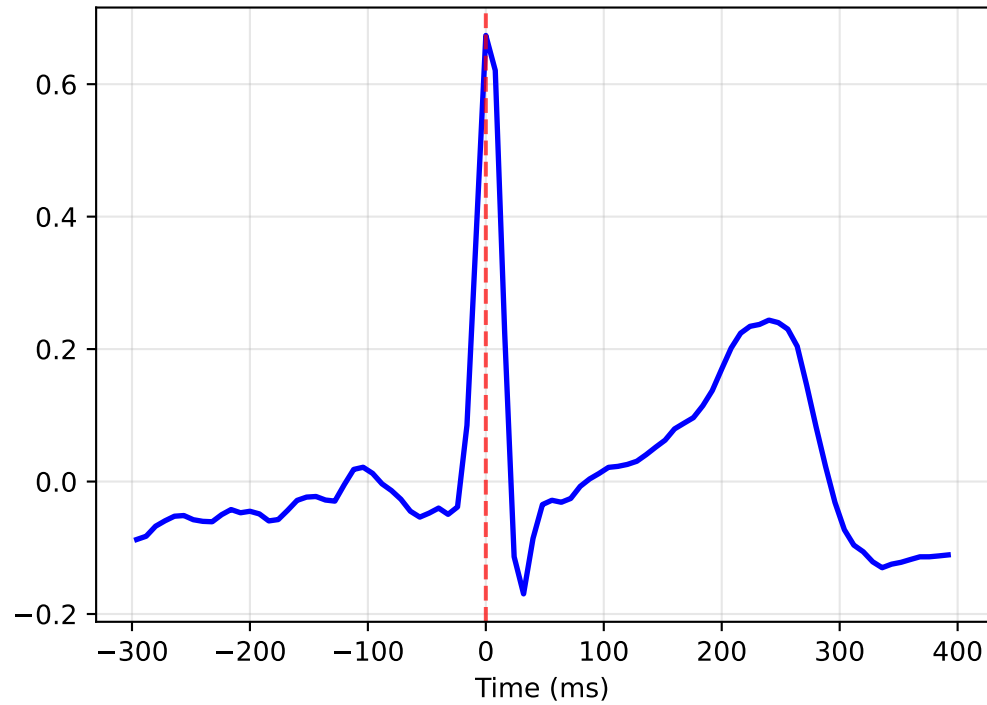


Cleaned ECG

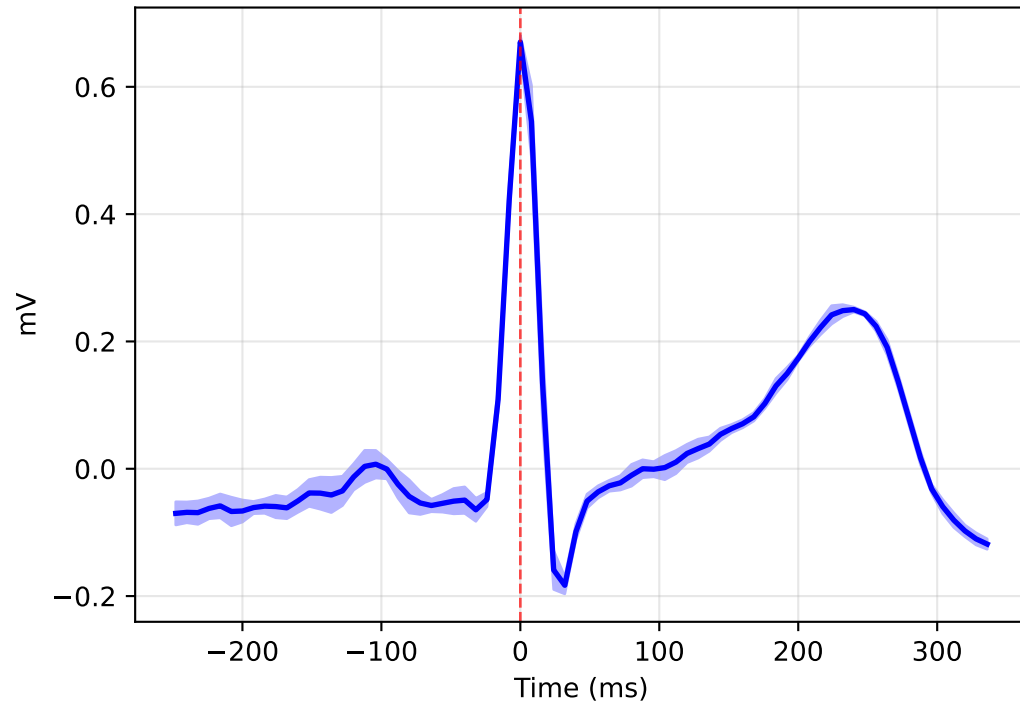


ECG Morphology - Sample1

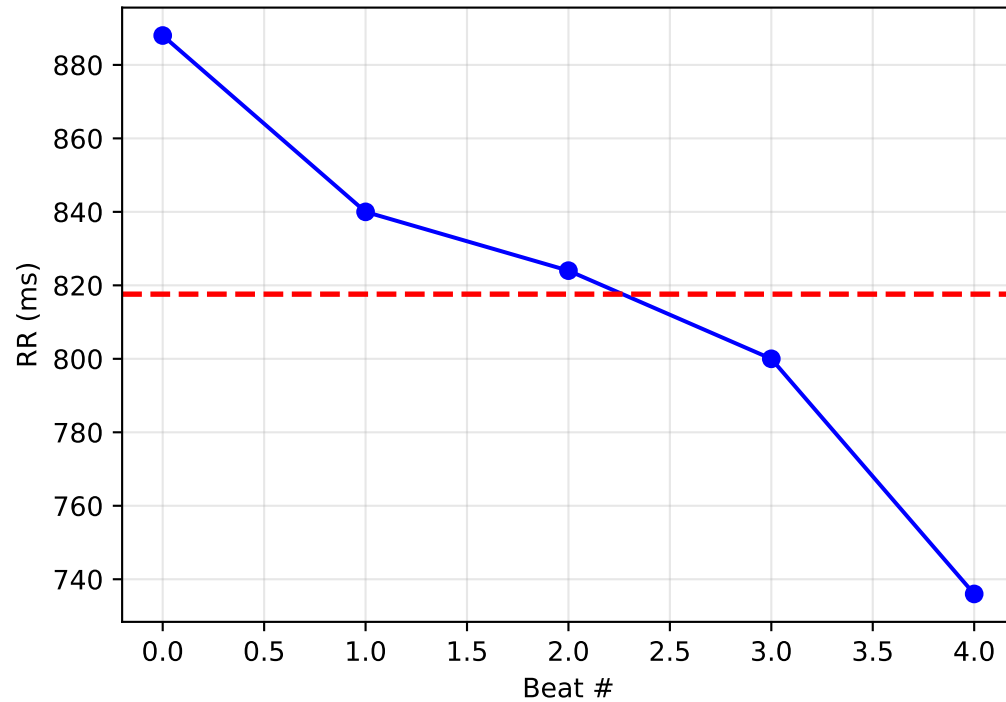
Single Beat



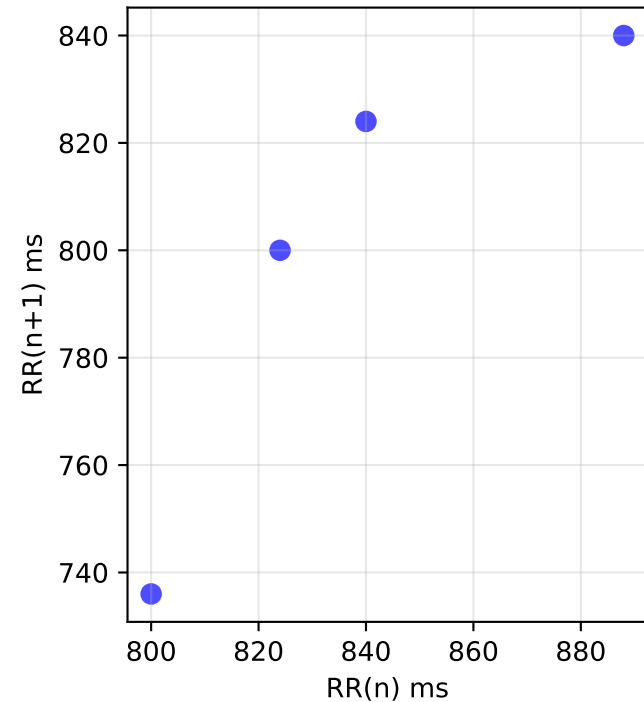
Average Beat (n=4)



RR Tachogram



Poincaré Plot



HRV Analysis - Sample1

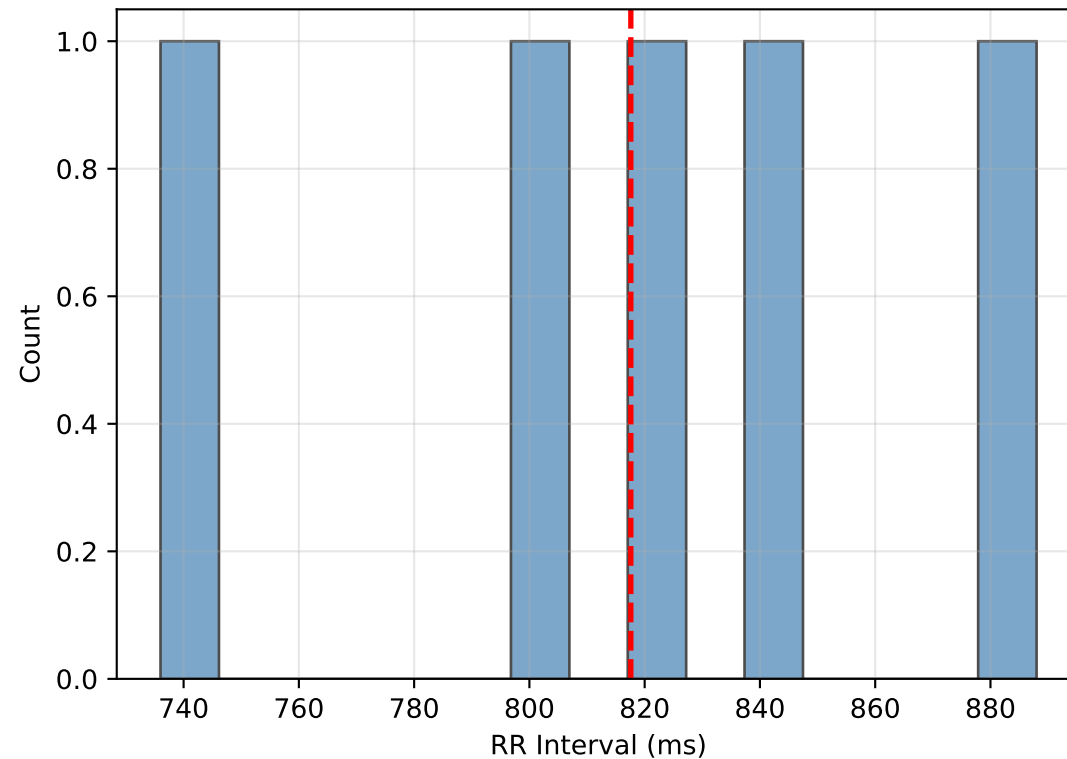
Time Domain

Average RR Interval:	817.60
Heart Rate Variability:	55.83
Short-term HRV:	42.52
Large Beat Changes %:	20.00
Median RR:	824.00

Frequency Domain

Requires 60+ seconds

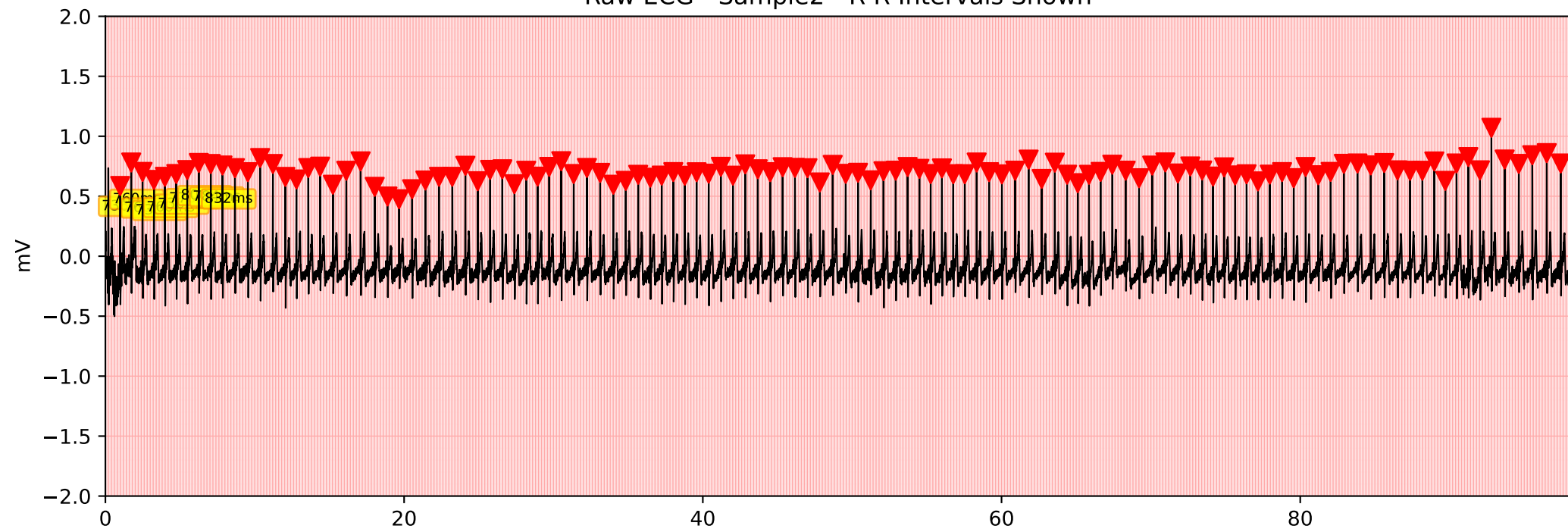
RR Distribution



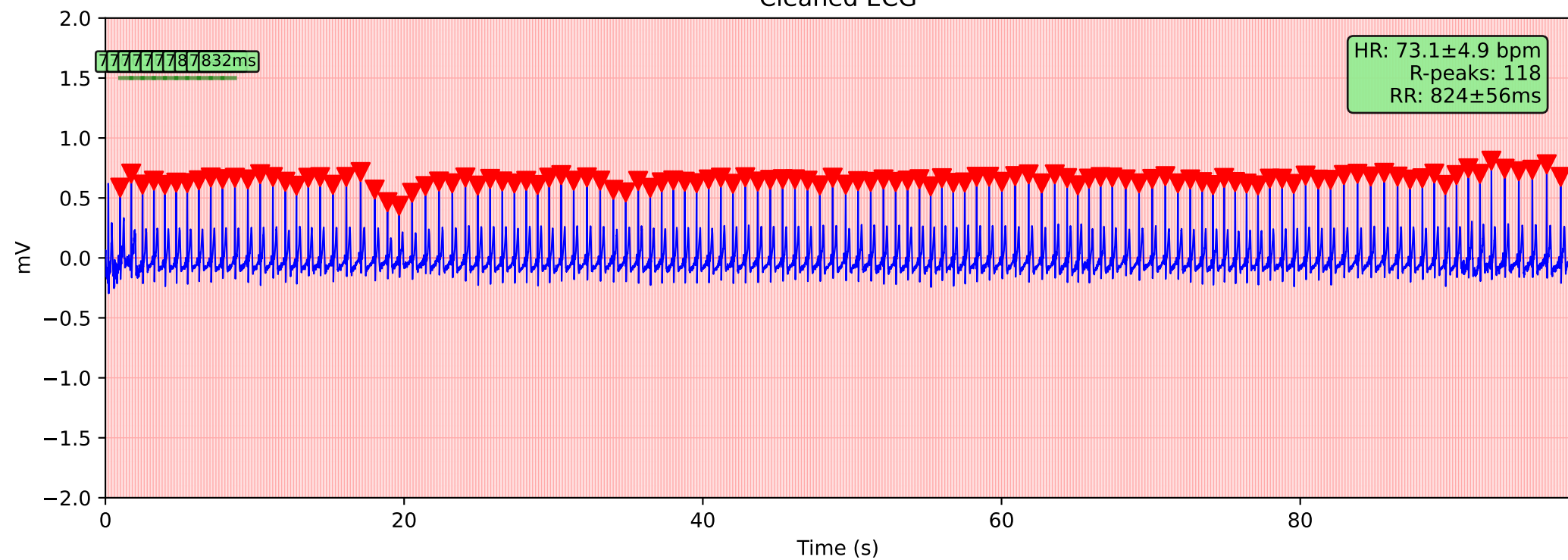
Nonlinear

ECG - Sample 2: Full Recording

Raw ECG - Sample2 - R-R Intervals Shown

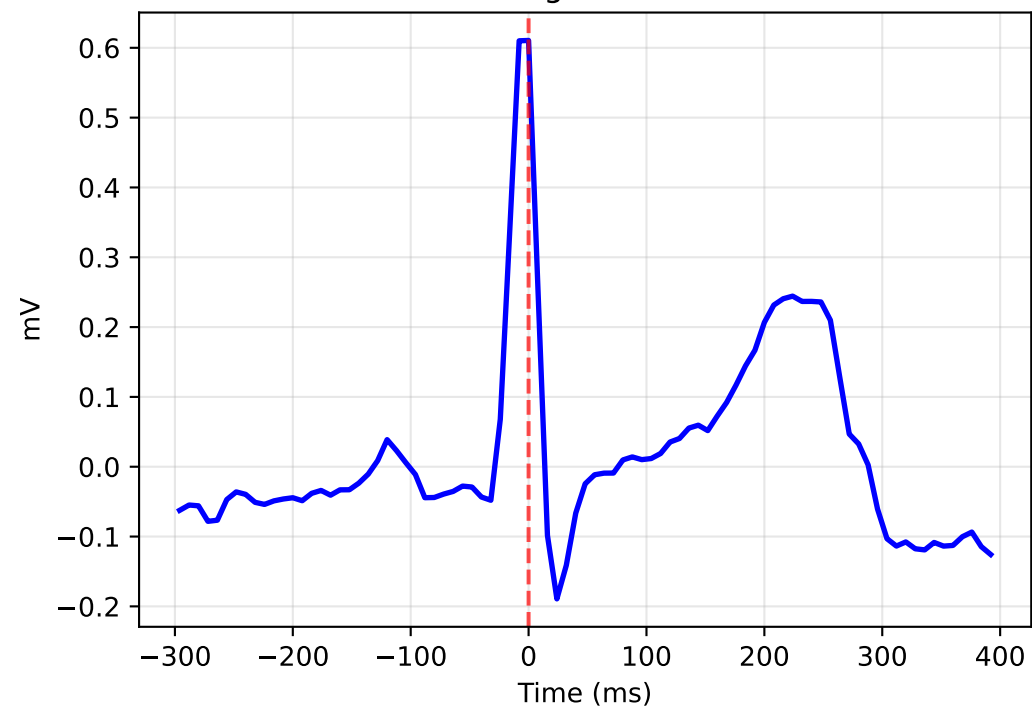


Cleaned ECG

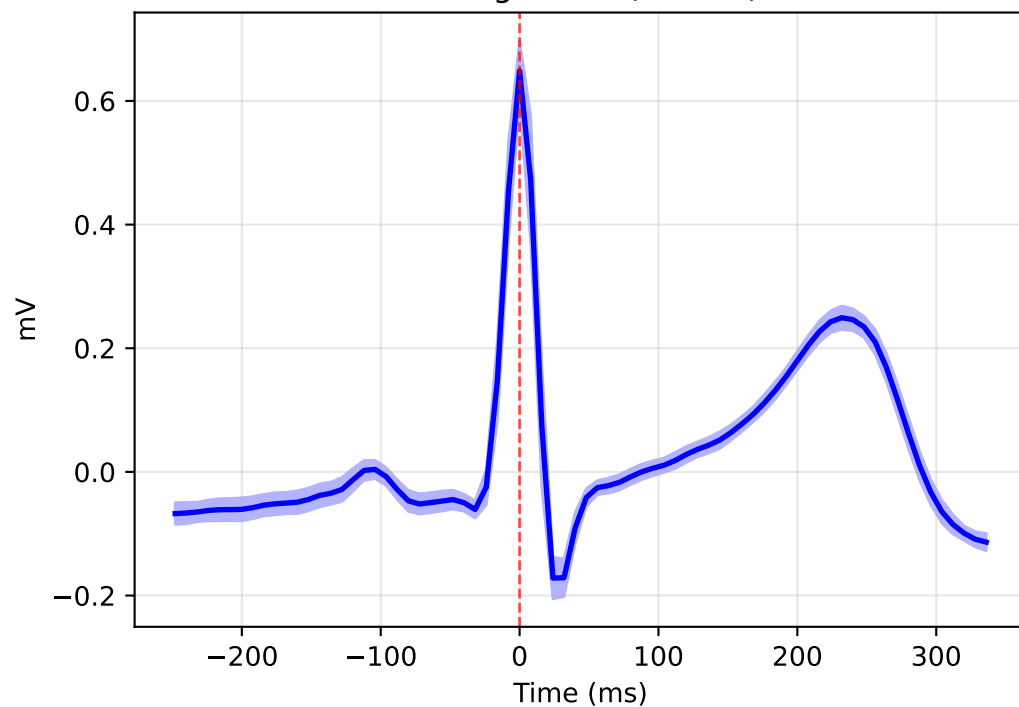


ECG Morphology - Sample2

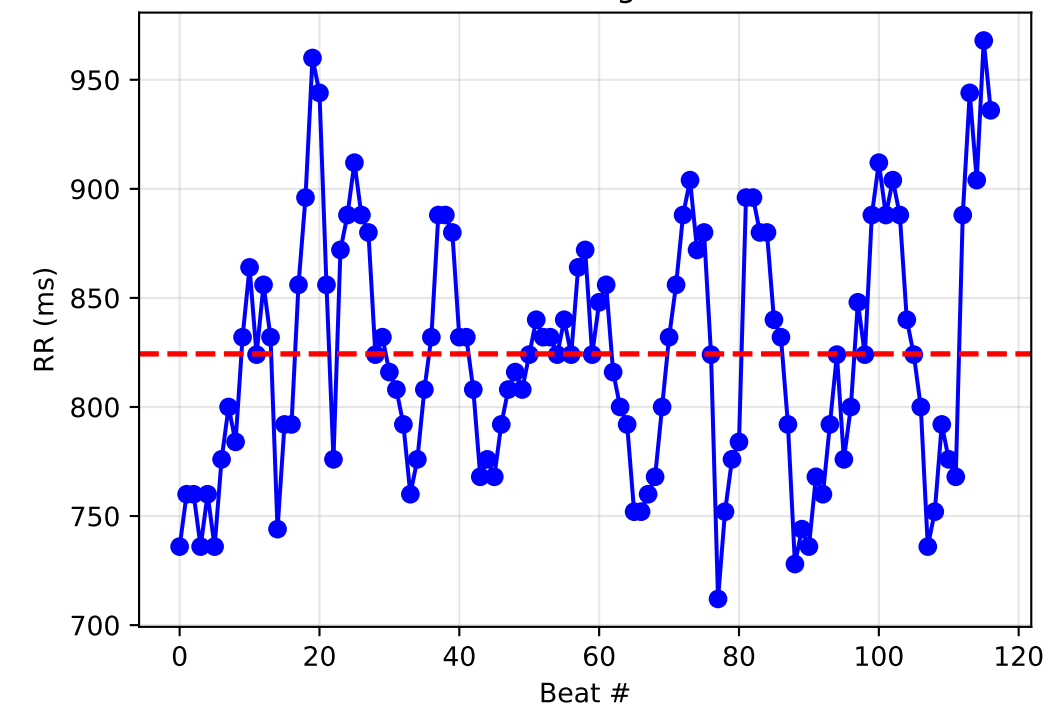
Single Beat



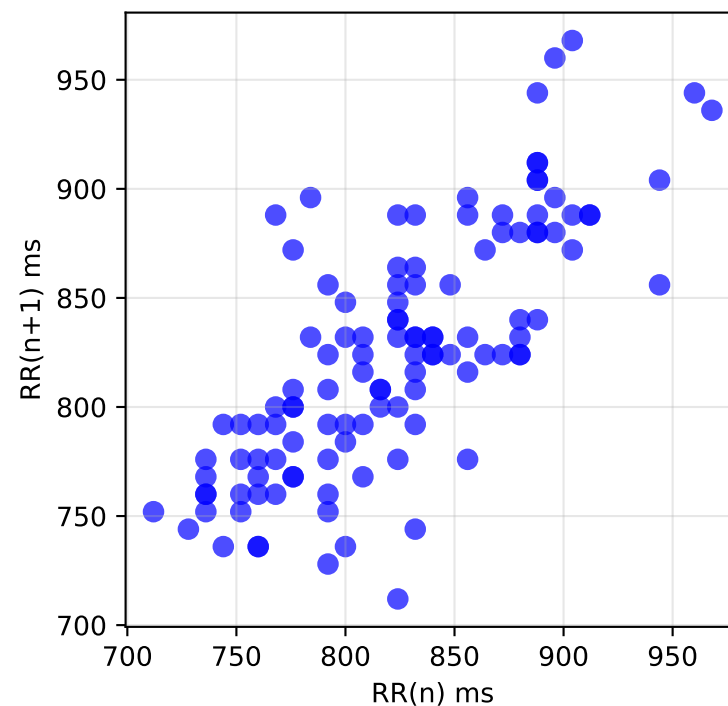
Average Beat (n=116)



RR Tachogram



Poincaré Plot



HRV Analysis - Sample2

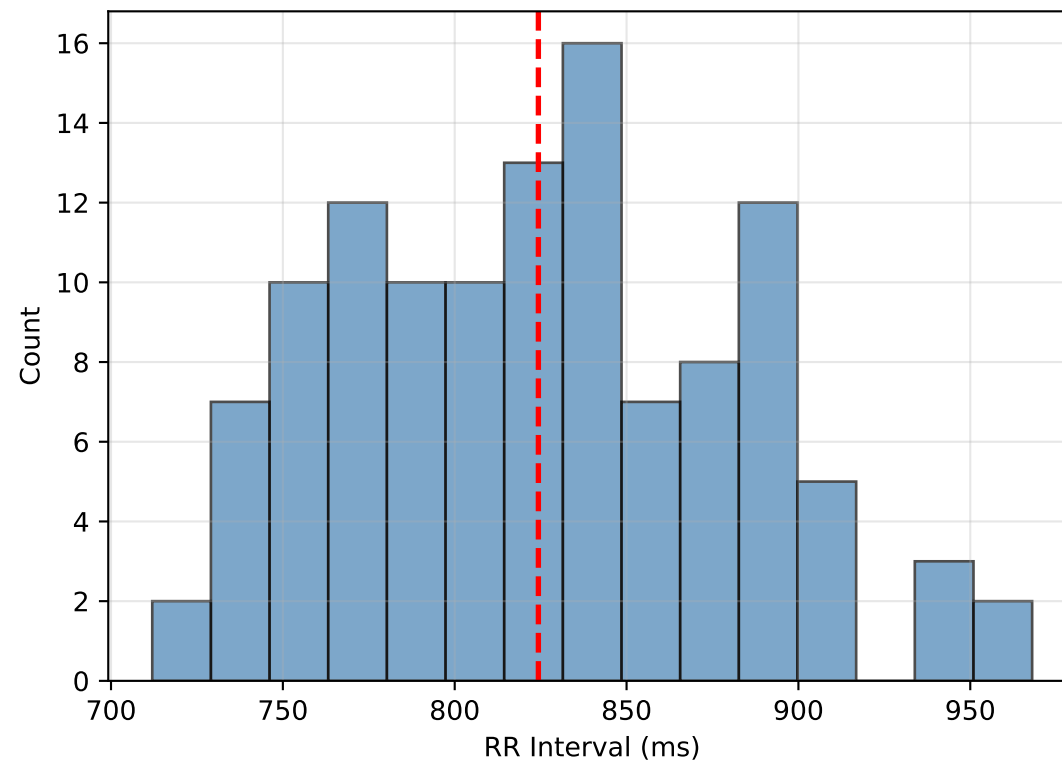
Time Domain

Average RR Interval:	824.34
Heart Rate Variability:	56.29
Short-term HRV:	38.59
Large Beat Changes %:	14.53
Median RR:	824.00

Frequency Domain

Low Freq Power:	0.08
High Freq Power:	0.01
LF/HF Ratio:	5.35

RR Distribution

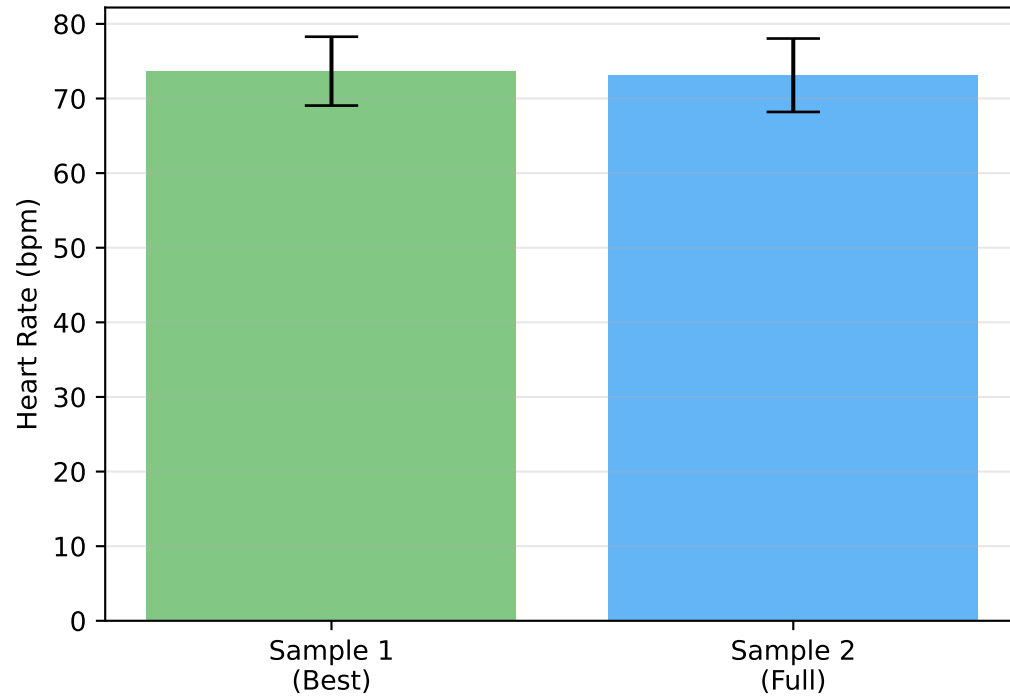


Nonlinear

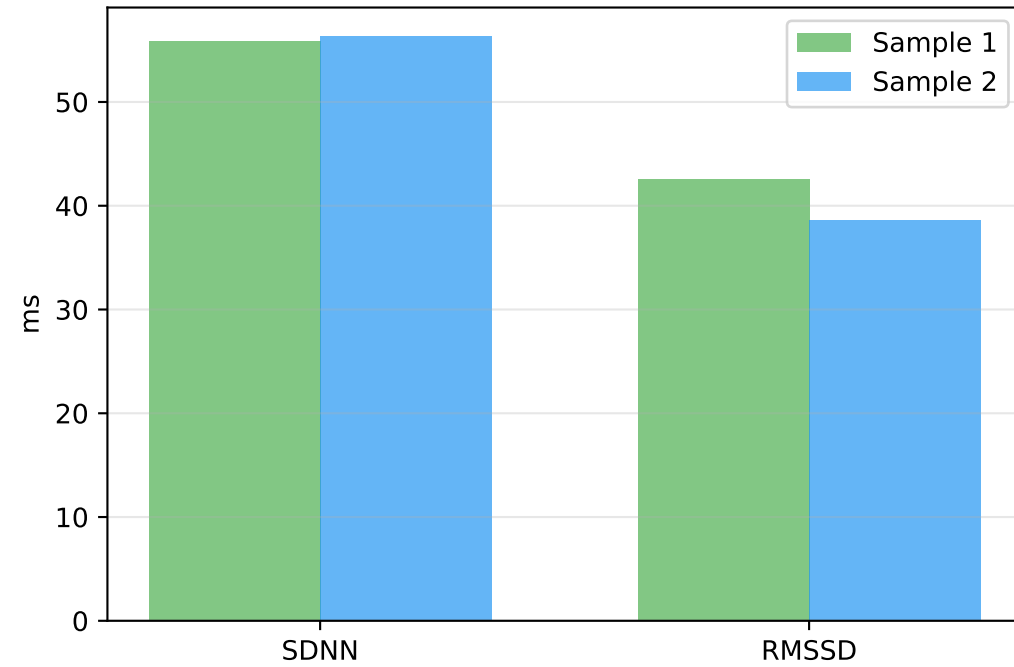
Poincaré SD1:	27.3775
Poincaré SD2:	73.9233
Approximate Entropy:	0.7477
DFA Short-term:	1.4680

ECG Comparison: Sample 1 vs Sample 2

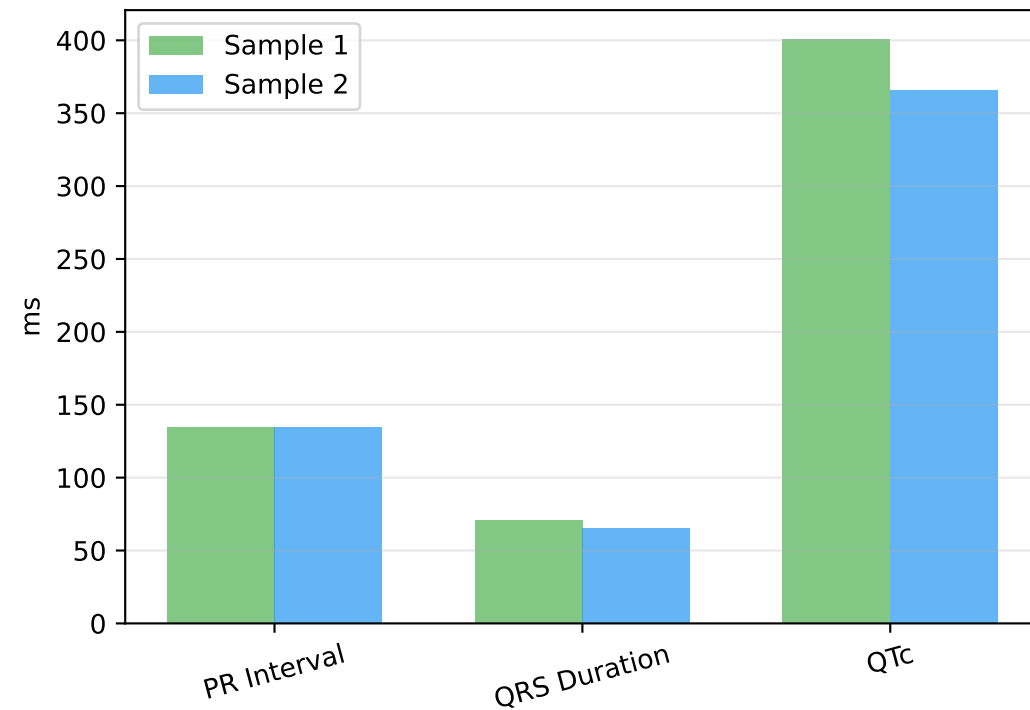
Heart Rate Comparison



HRV Time Domain



ECG Intervals



Key Differences

HR Difference: 0.5 bpm

SDNN Difference: 0.5 ms

Sample 1: 5.0s

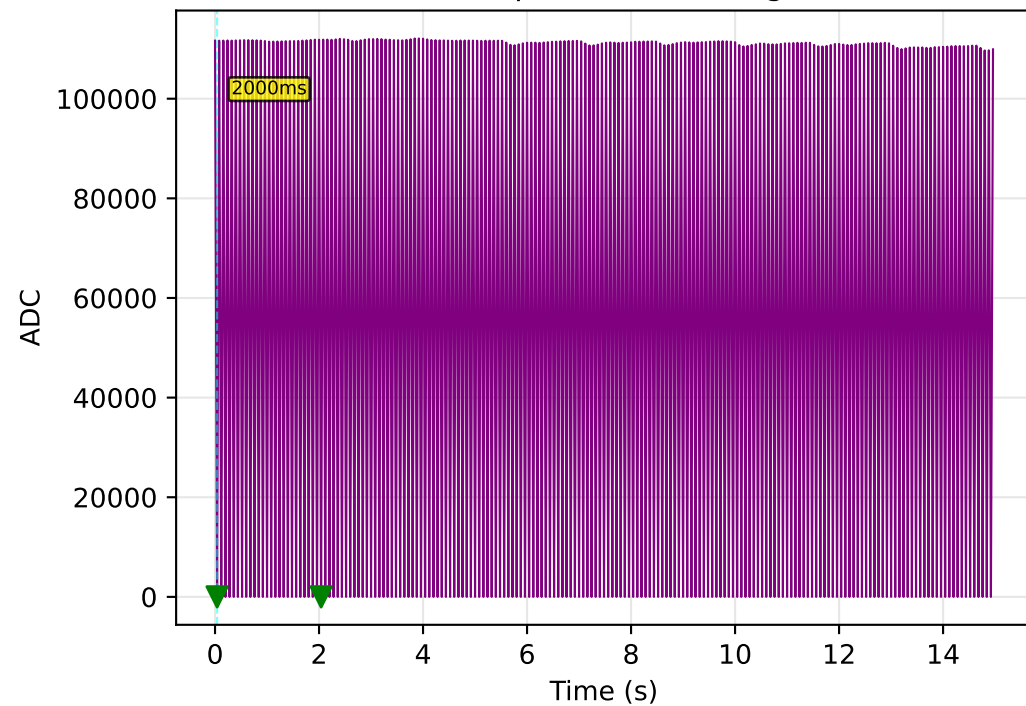
Sample 2: 98.1s

R-peaks Sample 1: 6

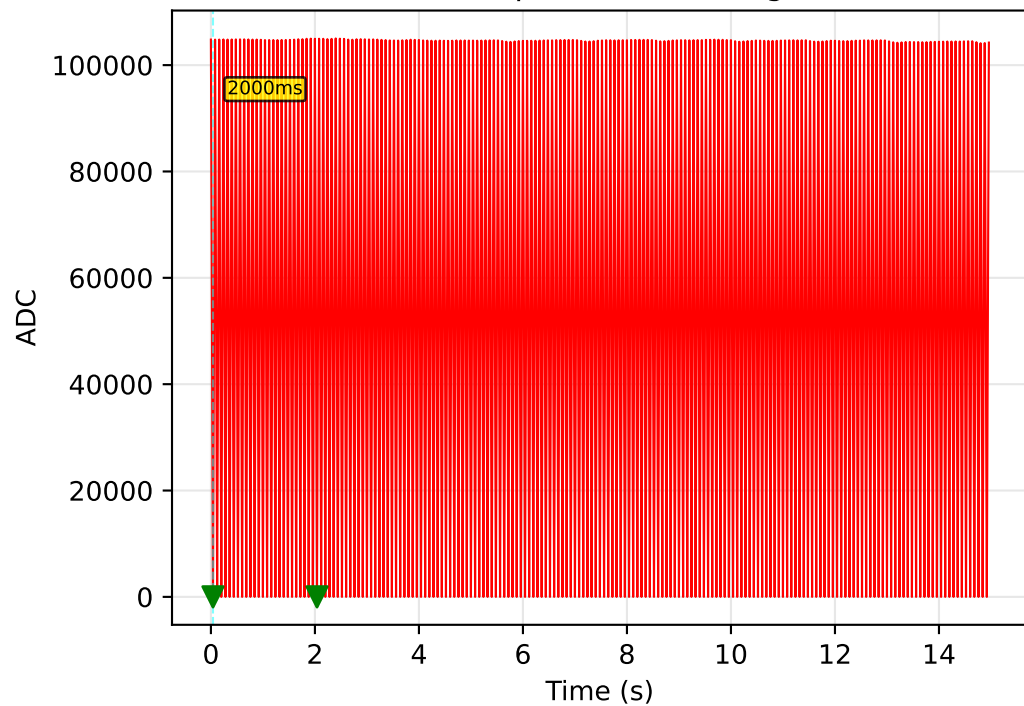
R-peaks Sample 2: 118

PPG Signals & Beat Segmentation - Sample1

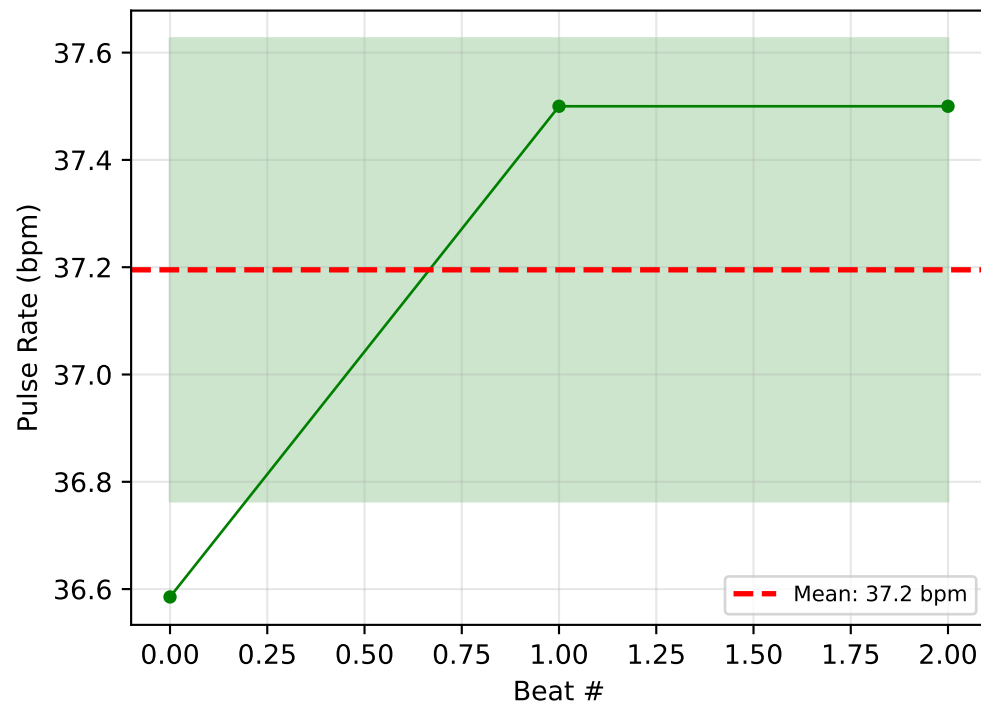
IR Channel - 10 peaks - Beat Segmentation



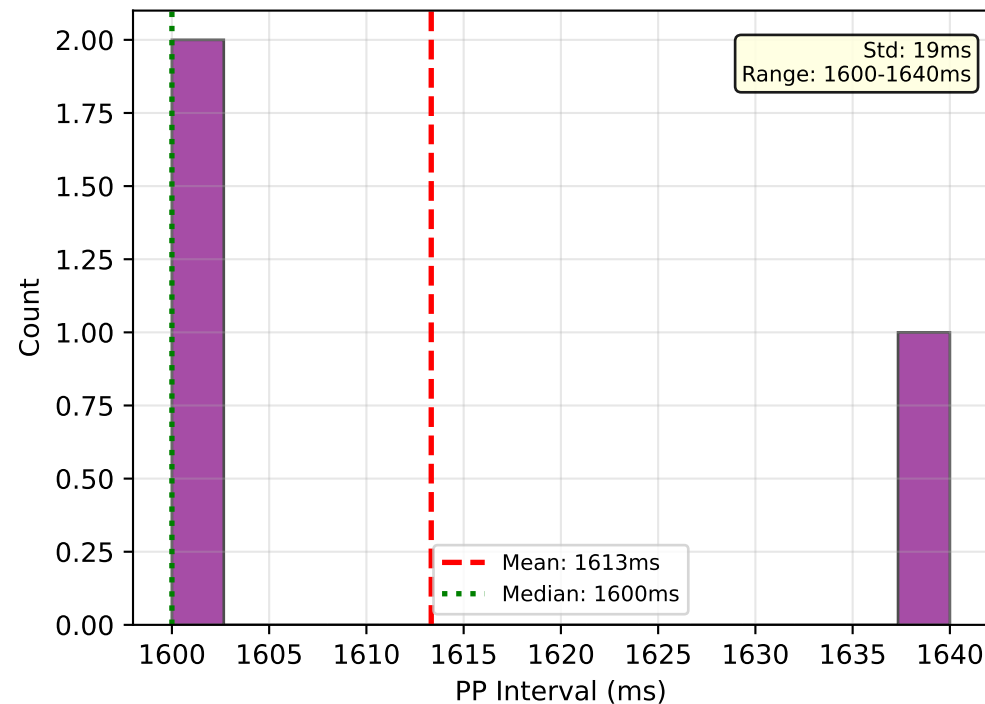
Red Channel - 3 peaks - Beat Segmentation



Pulse Rate from PPG

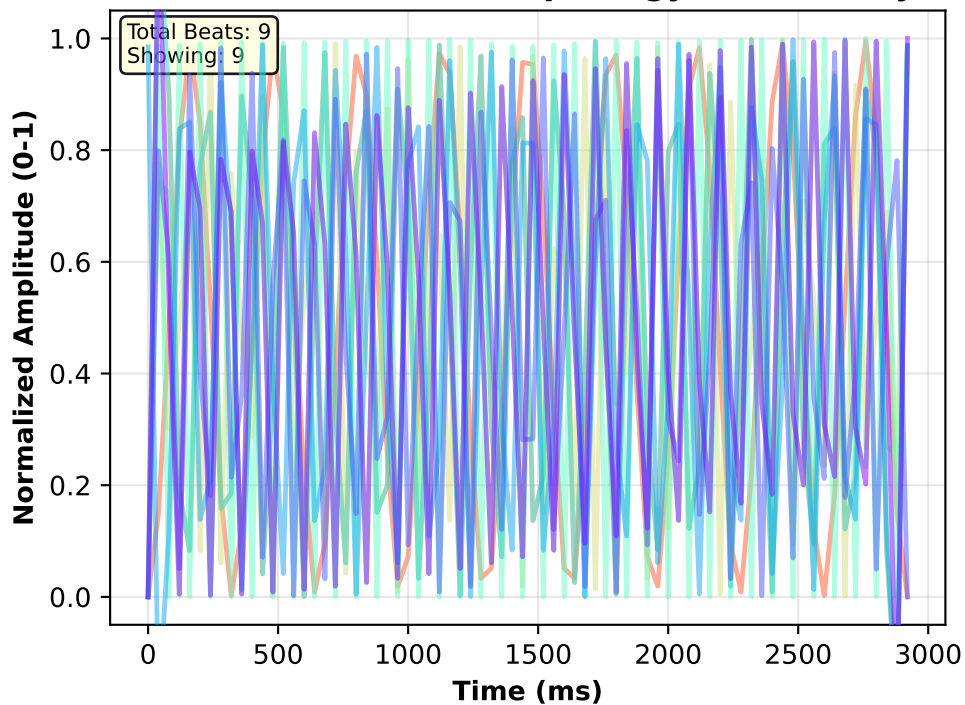


Peak-to-Peak Interval Distribution

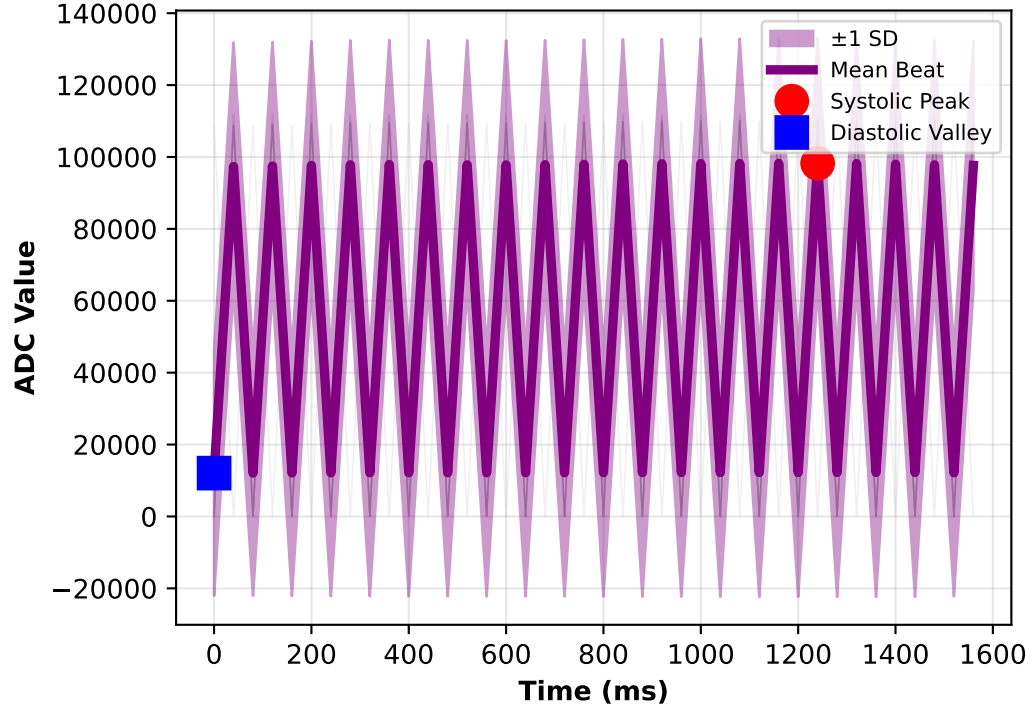


PPG Beat Morphology & Segmentation - Sample1

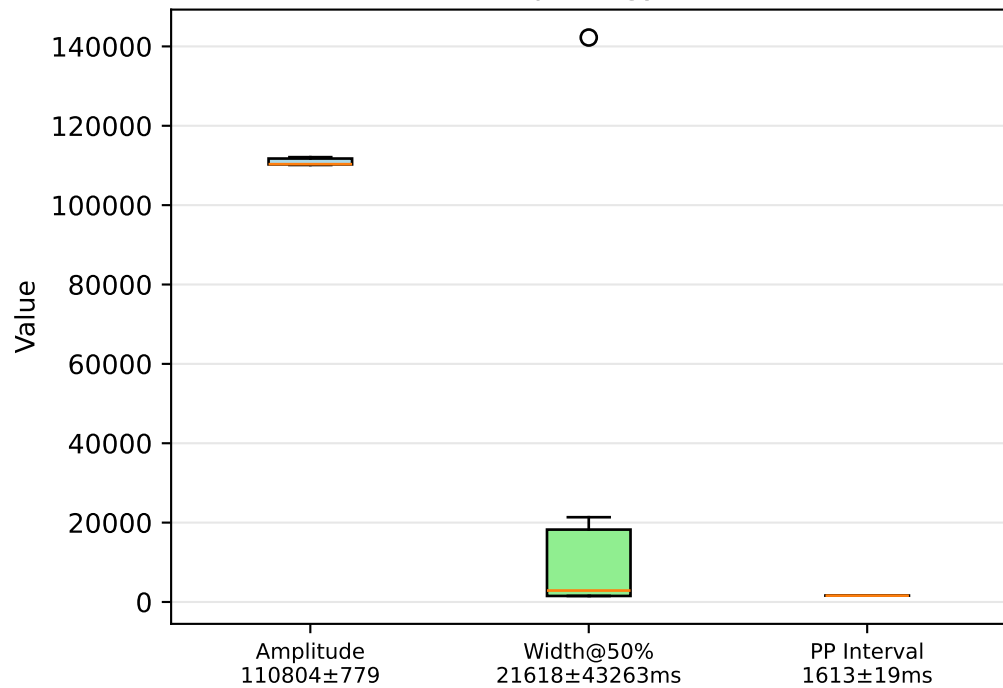
9 Beats Overlaid - Morphology Consistency



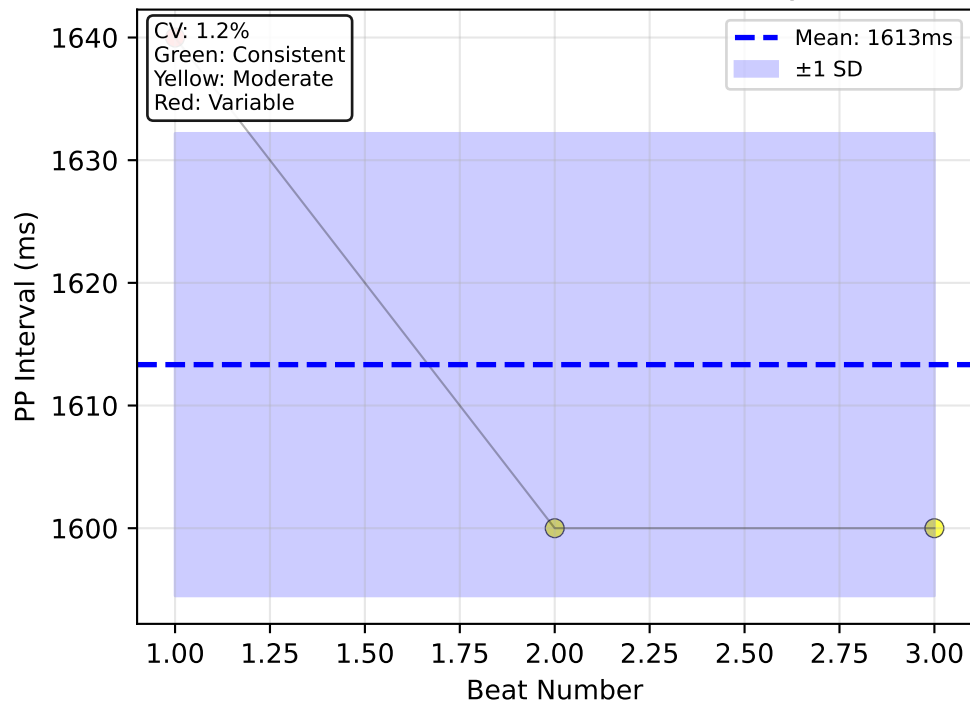
Mean Beat with All 9 Beats Overlaid



Beat Morphology Metrics



Beat-to-Beat Interval Variability



PPG Analysis - Sample1

Oxygen Saturation

81%

SpO2

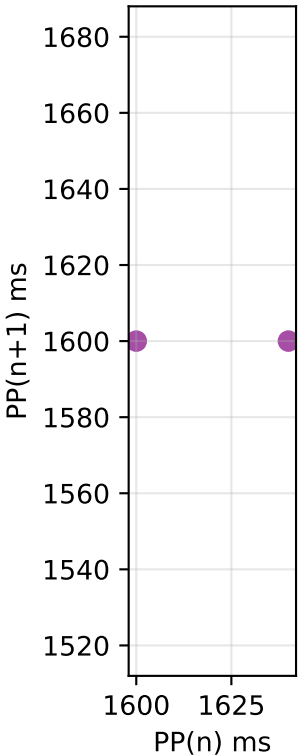
Signal Quality

Perfusion Index:	204.74%
Signal Quality:	1.000
Pulse Rate:	37.2 ± 0.4 bpm

PRV Time Domain

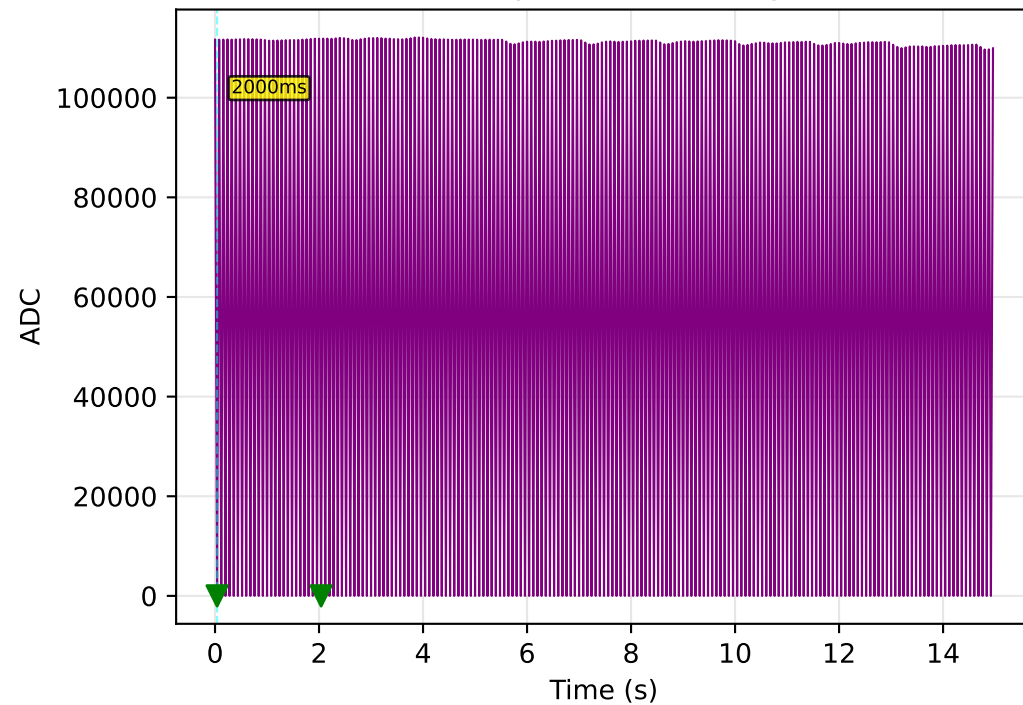
PRV_MeanNN:	21693.33
PRV_SDNN:	45872.69
PRV_RMSSD:	71069.56
PRV_pNN50:	77.78
PRV_pNN20:	88.89

PPG Poincaré Plot

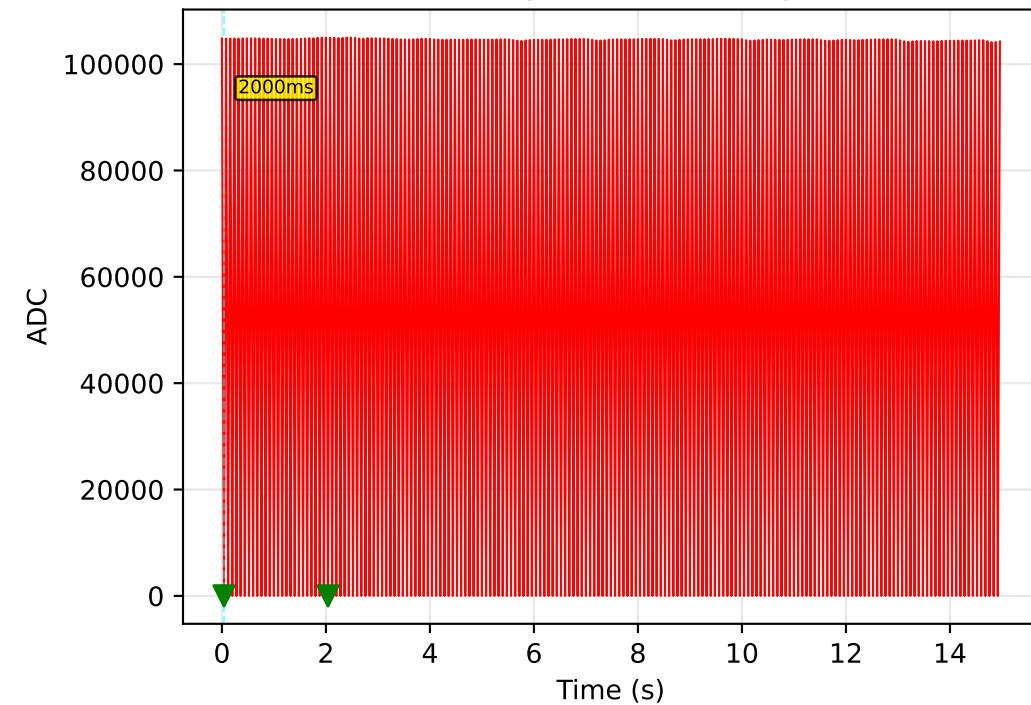


PPG Signals & Beat Segmentation - Sample2

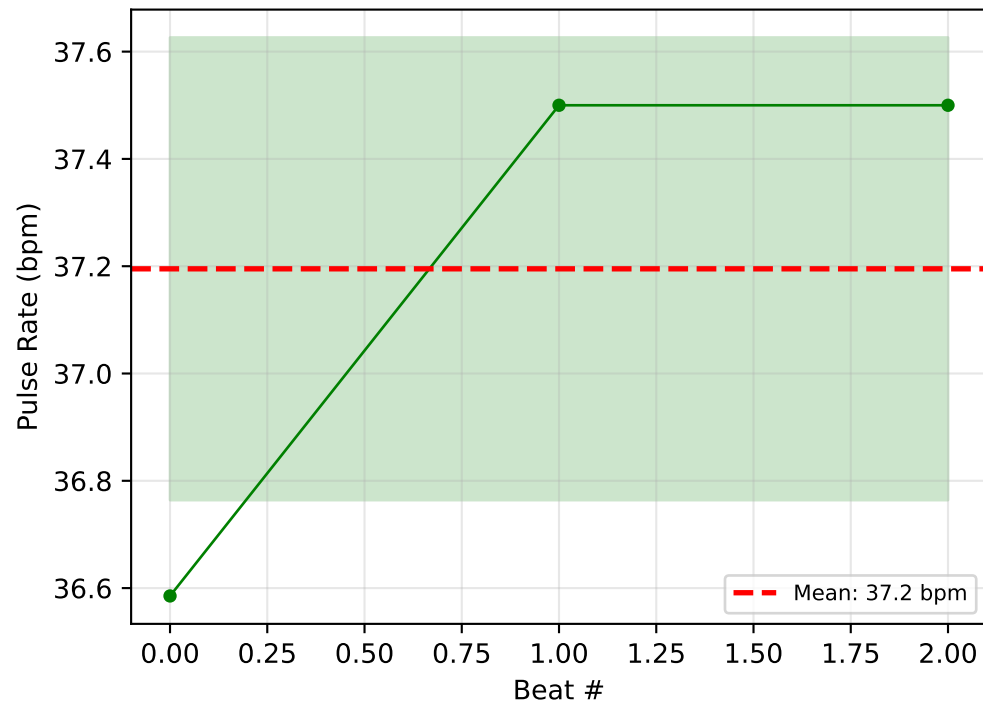
IR Channel - 10 peaks - Beat Segmentation



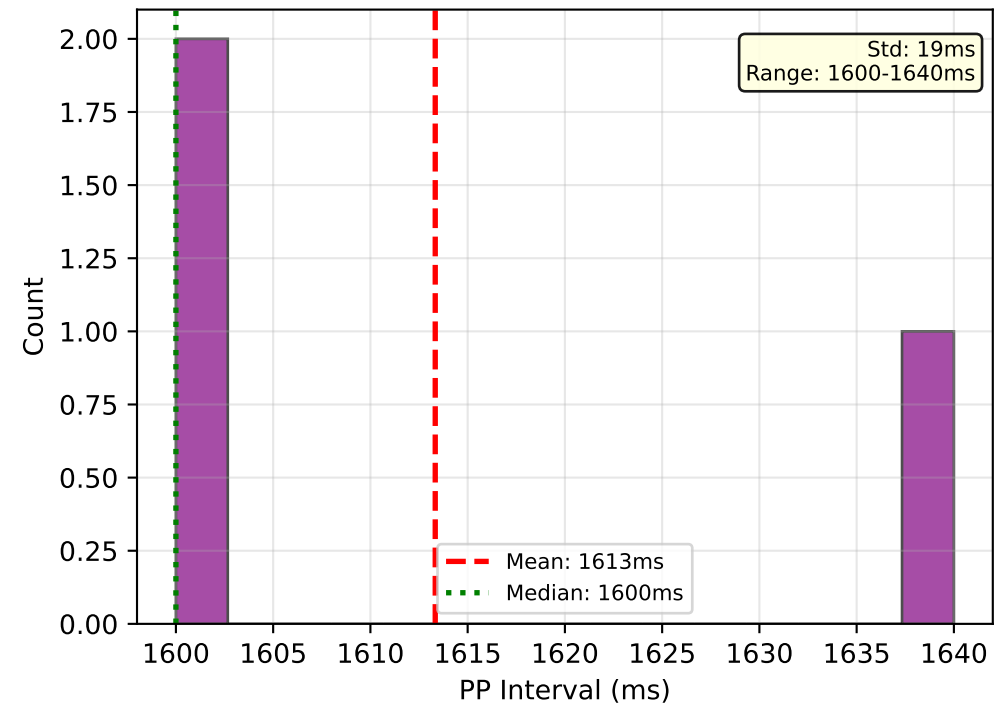
Red Channel - 3 peaks - Beat Segmentation



Pulse Rate from PPG

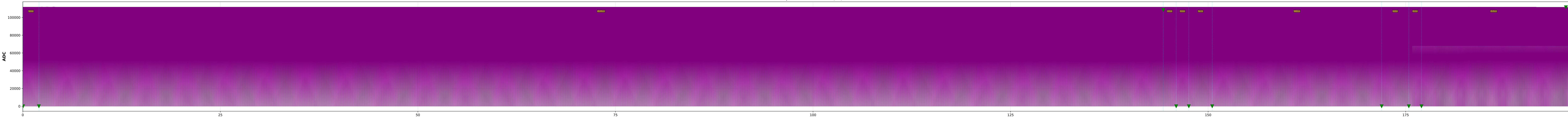


Peak-to-Peak Interval Distribution

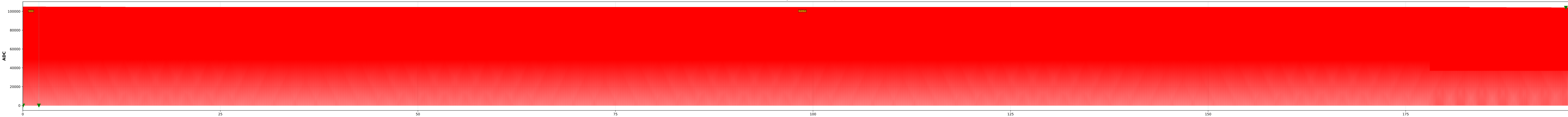


PPG Signals - Sample2 - FULL SCROLLABLE VIEW

IR Channel - 10 peaks - All PP Intervals (ms)



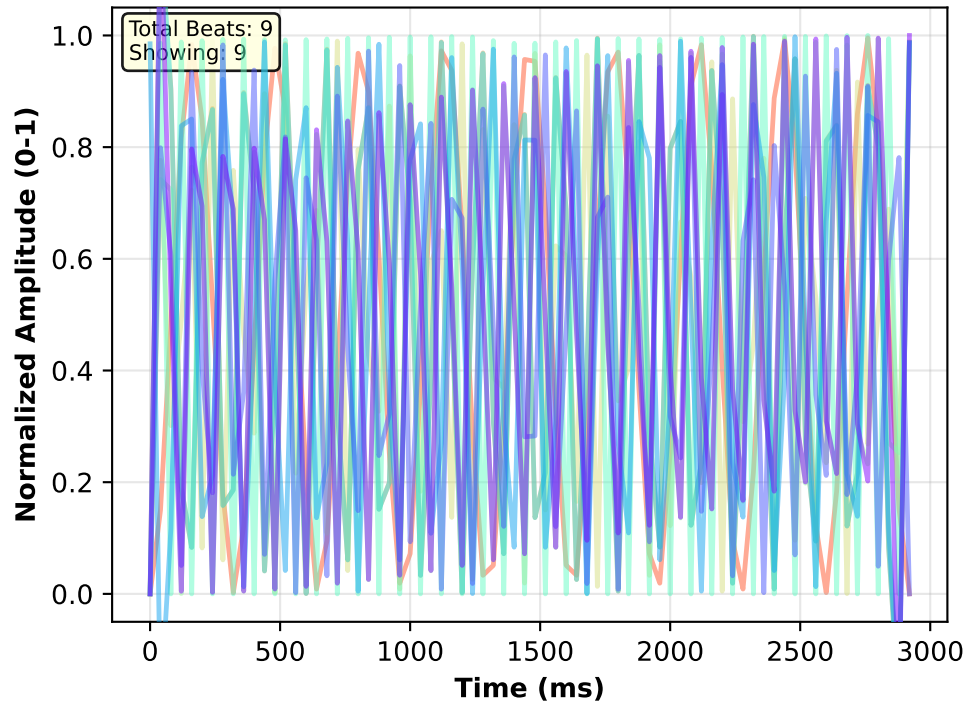
Red Channel - 3 peaks - All PP Intervals (ms)



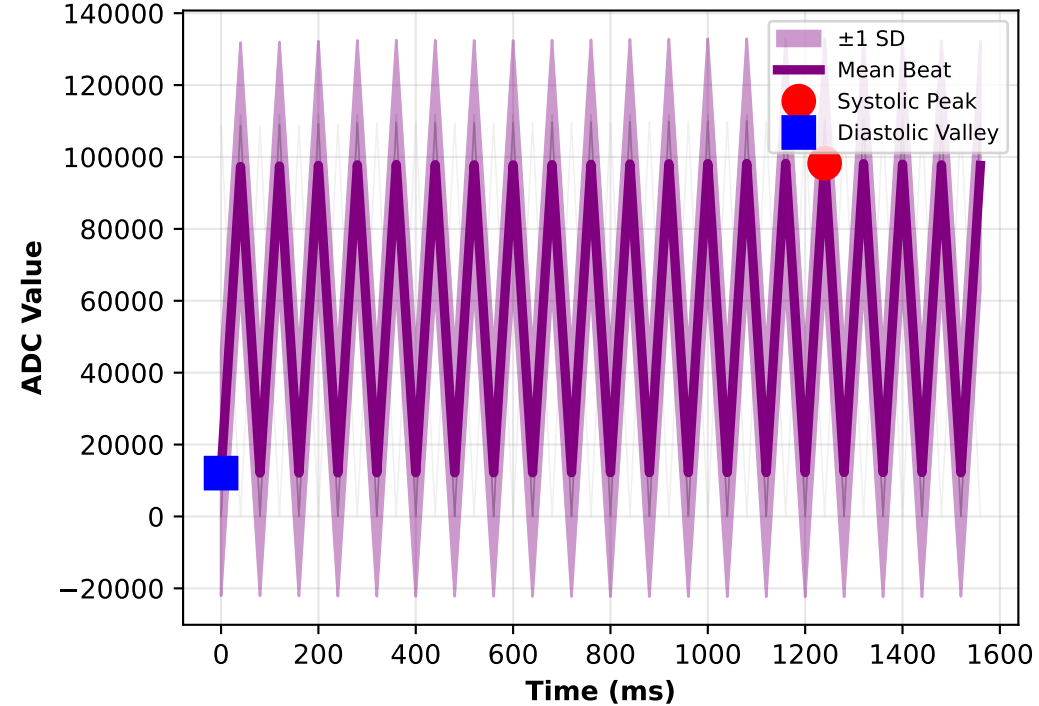
Time (s)

PPG Beat Morphology & Segmentation - Sample2

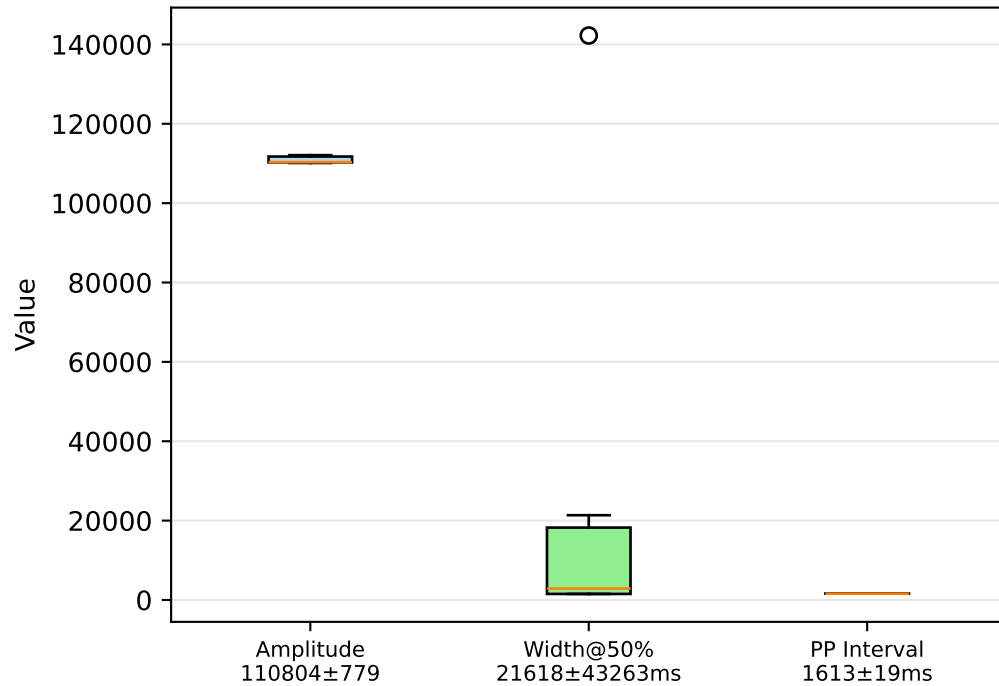
9 Beats Overlaid - Morphology Consistency



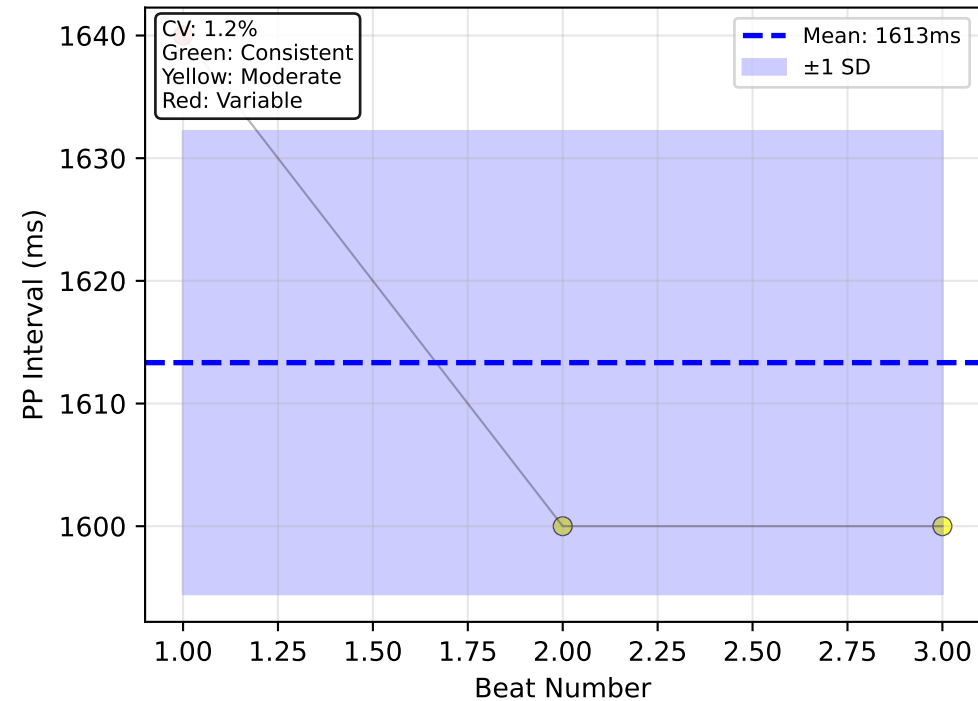
Mean Beat with All 9 Beats Overlaid



Beat Morphology Metrics



Beat-to-Beat Interval Variability



PPG Analysis - Sample2

Oxygen Saturation

81%

SpO2

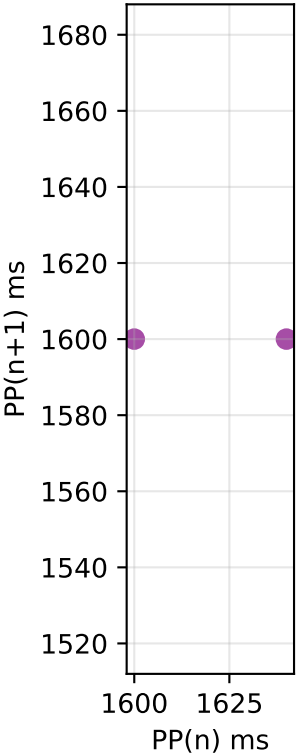
Signal Quality

Perfusion Index: 204.74%
Signal Quality: 1.000
Pulse Rate: 37.2 ± 0.4 bpm

PRV Time Domain

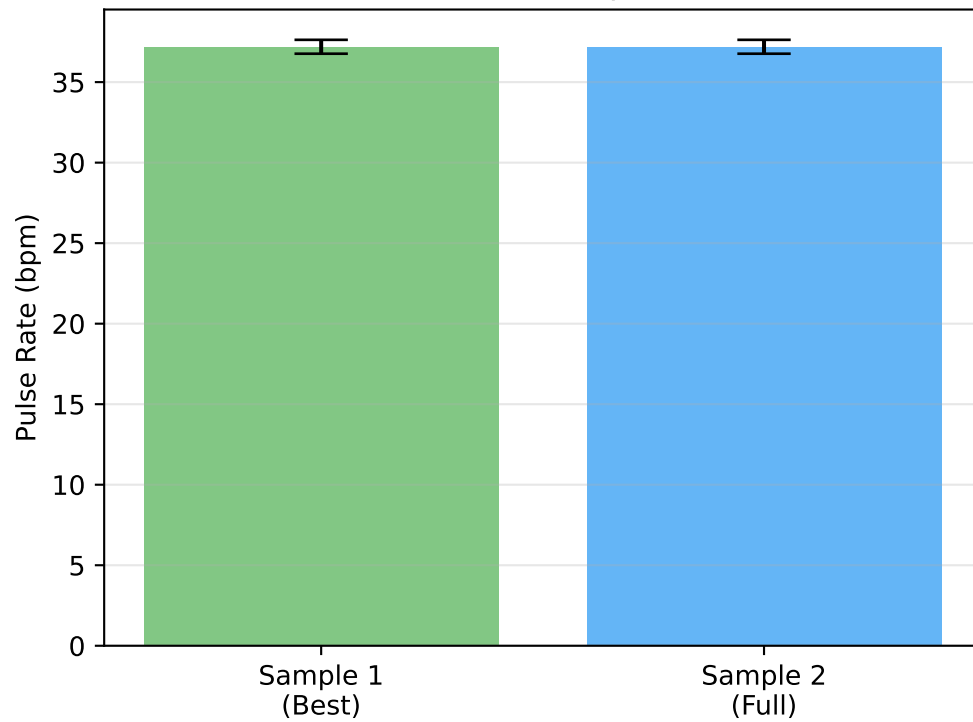
PRV_MeanNN: 21693.33
PRV_SDNN: 45872.69
PRV_RMSSD: 71069.56
PRV_pNN50: 77.78
PRV_pNN20: 88.89

PPG Poincaré Plot

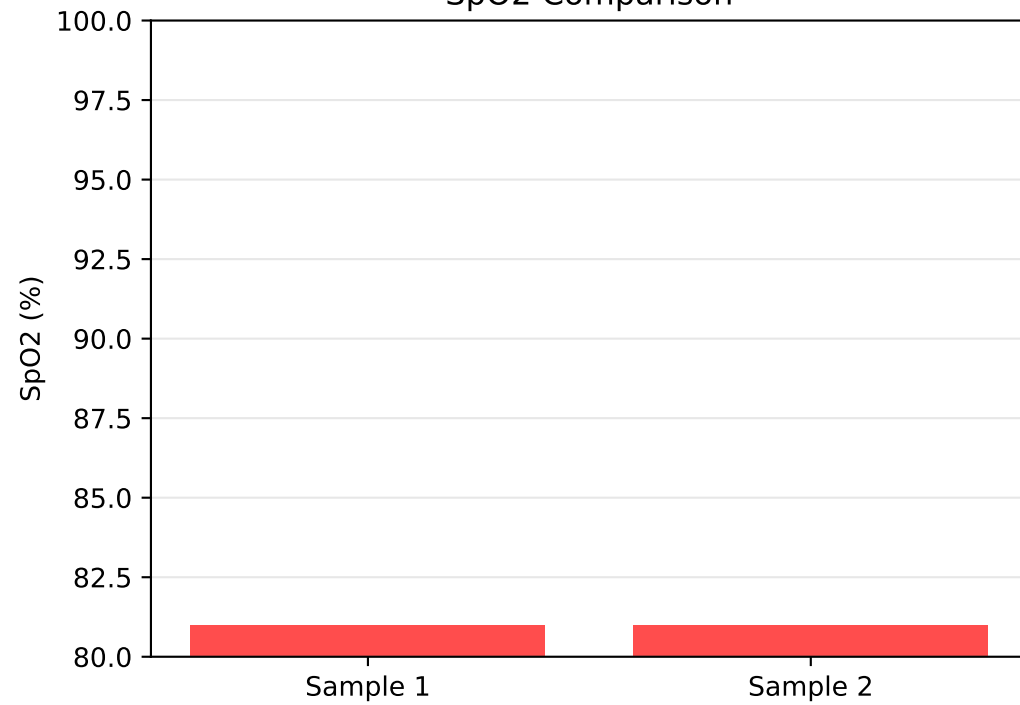


PPG Comparison: Sample 1 vs Sample 2

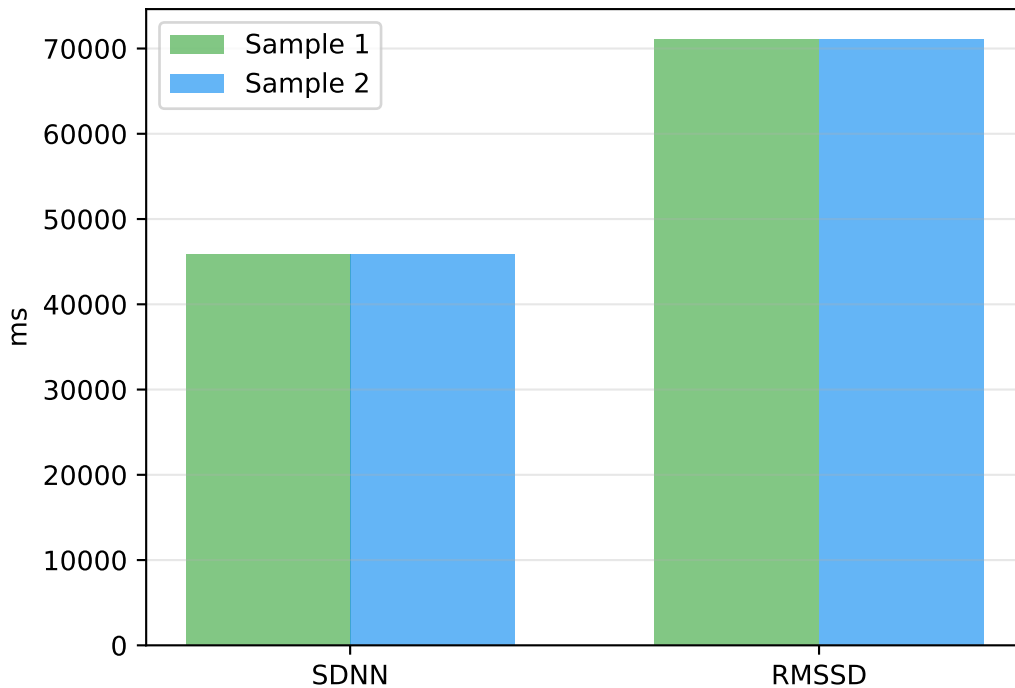
Pulse Rate Comparison



SpO2 Comparison



PRV Comparison



Key Differences

Pulse Rate Diff: 0.0 bpm

SpO2 Difference: 0%

Sample 1: 196.2s

Sample 2: 196.2s

IR peaks Sample 1: 10

IR peaks Sample 2: 10

Summary - All Samples

ECG Sample 1 (Best 5s Window - Basic Analysis):

Duration:	5.0s
R-peaks:	6
Analysis Type:	Time-domain only
Heart Rate:	73.7 ± 4.6 bpm

ECG Sample 2 (Full Recording - Complete Analysis):

Duration:	98.1s
R-peaks:	118
Analysis Type:	Time + Freq + Nonlinear
Heart Rate:	73.1 ± 4.9 bpm
Long-term HRV:	✓ Available

PPG Sample 1 (Full Recording - Basic Analysis):

Duration:	196.2s
IR Peaks:	10
Analysis Type:	Time-domain only
SpO2:	81%

PPG Sample 2 (Full Recording - Complete Analysis):

Duration:	196.2s
IR Peaks:	10
Analysis Type:	Time + Freq + Nonlinear
SpO2:	81%

Understanding Your Results

- **Oxygen Saturation:**

Blood oxygen. Normal: 95-100%.

- **Heart Rate Variability:**

How much heartbeat timing varies. Higher = healthier. Normal: 50-100ms.

- **Short-term HRV:**

Beat-to-beat changes. Normal: 20-50ms.

- **Pulse Rate Variability:**

PRV from PPG.